有理函数积分表

以下是部分有理函数的积分表。

$$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C \qquad (n \neq -1)$$

$$\int \frac{1}{ax+b} dx = \frac{1}{a} \ln|ax+b| + C$$

$$\int x(ax+b)^n dx = \frac{a(n+1)x-b}{a^2(n+1)(n+2)}(ax+b)^{n+1} + C \qquad (n \notin \{1,2\})$$

$$\int \frac{x}{ax+b} dx = \frac{x}{a} - \frac{b}{a^2} \ln|ax+b| + C$$

$$\int \frac{x}{(ax+b)^2} dx = \frac{b}{a^2(ax+b)} + \frac{1}{a^2} \ln|ax+b| + C$$

$$\int \frac{x}{(ax+b)^n} dx = \frac{a(1-n)x-b}{a^2(n-1)(n-2)(ax+b)^{n-1}} + C \qquad (n \notin \{1,2\})$$

$$\int \frac{x^2}{ax+b} dx = \frac{1}{a^3}\left[\frac{(ax+b)^2}{2} - 2b(ax+b) + b^2 \ln|ax+b|\right] + C$$

$$\int \frac{x^2}{(ax+b)^2} dx = \frac{1}{a^3}\left(ax+b-2b\ln|ax+b| - \frac{b^2}{ax+b}\right) + C$$

$$\int \frac{x^2}{(ax+b)^3} dx = \frac{1}{a^3}\left[\ln|ax+b| + \frac{2b}{ax+b} - \frac{b^2}{2(ax+b)^2}\right] + C$$

$$\int \frac{x^2}{(ax+b)^n} dx = \frac{1}{a^3}\left[-\frac{1}{(n-3)(ax+b)^{n-3}} + \frac{2b}{(n-2)(ax+b)^{n-2}} - \frac{b^2}{(n-1)(ax+b)}\right.$$

$(n \notin \{1,2,3\})$

$$\int \frac{dx}{x(ax+b)} = -\frac{1}{b} \ln\left|\frac{ax+b}{x}\right| + C$$

$$\int \frac{dx}{x^2(ax+b)} = -\frac{1}{bx} + \frac{a}{b^2} \ln\left|\frac{ax+b}{x}\right| + C$$

$$\int \frac{dx}{x^2(ax+b)^2} = -a\left[\frac{1}{b^2(ax+b)} + \frac{1}{ab^2x} - \frac{2}{b^3} \ln\left|\frac{ax+b}{x}\right|\right] + C$$

$$\int \frac{dx}{x^2+a^2} = \frac{1}{a}\arctan\frac{x}{a} + C$$

$$\int \frac{dx}{x^2-a^2} = -\frac{1}{a}\operatorname{artanh}\frac{x}{a} = \frac{1}{2a}\ln\frac{a-x}{a+x} + C \qquad (|x|<|a|)$$

$$\int \frac{dx}{x^2-a^2} = -\frac{1}{a}\operatorname{arcoth}\frac{x}{a} = \frac{1}{2a}\ln\frac{x-a}{x+a} + C \qquad (|x|>|a|)$$

$$\int \frac{dx}{ax^2+bx+c} = \frac{2}{\sqrt{4ac-b^2}}\arctan\frac{2ax+b}{\sqrt{4ac-b^2}} + C, (4ac-b^2>0)$$

$$\int \frac{dx}{ax^2+bx+c} = \frac{2}{\sqrt{b^2-4ac}}\operatorname{artanh}\frac{2ax+b}{\sqrt{b^2-4ac}} + C = \frac{1}{\sqrt{b^2-4ac}}\ln\left|\frac{2ax+b-\sqrt{b^2}}{2ax+b+\sqrt{b^2}}\right.$$

$$\int \frac{dx}{ax^2+bx+c} = -\frac{2}{2ax+b} + C \qquad (4ac-b^2=0)$$

$$\int \frac{x}{ax^2+bx+c}dx = \frac{1}{2a}\ln|ax^2+bx+c| - \frac{b}{2a}\int\frac{dx}{ax^2+bx+c} + C$$

$$\int \frac{mx+n}{ax^2+bx+c}dx = \frac{m}{2a}\ln|ax^2+bx+c| + \frac{2an-bm}{a\sqrt{4ac-b^2}}\arctan\frac{2ax+b}{\sqrt{4ac-b^2}} + C \qquad ($$

$$\int \frac{mx+n}{ax^2+bx+c}dx = \frac{m}{2a}\ln|ax^2+bx+c| - \frac{2an-bm}{a\sqrt{b^2-4ac}}\operatorname{artanh}\frac{2ax+b}{\sqrt{b^2-4ac}} + C \qquad ($$

$$\int \frac{mx+n}{ax^2+bx+c}dx = \frac{m}{2a}\ln|ax^2+bx+c| - \frac{2an-bm}{a(2ax+b)} + C \qquad (4ac-b^2=0)$$

$$\int \frac{dx}{(ax^2+bx+c)^n} = \frac{2ax+b}{(n-1)(4ac-b^2)(ax^2+bx+c)^{n-1}} + \frac{(2n-3)2a}{(n-1)(4ac-b^2)}\int\frac{}{(ax}$$

$$\int \frac{x}{(ax^2+bx+c)^n}dx = \frac{bx+2c}{(n-1)(4ac-b^2)(ax^2+bx+c)^{n-1}} - \frac{b(2n-3)}{(n-1)(4ac-b^2)}\int$$

对于任意的有理函数，我们都能通过部分分式(partial fraction)把该函数分拆为数个函数的总和，其中每个函数符合以下的形式：$\frac{px+q}{(ax^2+bx+c)^n}$。我们继而能把每一个该种形式的函数作积分运算：

$$\int \frac{dx}{x(ax^2+bx+c)} = \frac{1}{2c}\ln\left|\frac{x^2}{ax^2+bx+c}\right| - \frac{b}{2c}\int \frac{dx}{ax^2+bx+c} + C$$
